

Review

Thermal Runaway in Lithium-Ion Batteries: A Review of Mechanisms, Prediction Approaches, and Mitigation Strategies

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Abstract

Thermal runaway is one of the most critical safety challenges limiting the widespread deployment of lithium-ion batteries in electric vehicles, energy storage systems, and aerospace applications. With the continuous increase in battery energy density, the fault-to-failure transition becomes increasingly rapid, which makes early detection and effective intervention quite difficult. This review systematically summarizes the fundamental mechanisms underlying thermal runaway that drive the escalation of battery hazards. Existing thermal runaway prediction and early warning approaches are comprehensively classified into electrical, thermal, mechanical/gas, and data-driven categories. The detection principles, performance characteristics, and current limitations are critically analyzed. Furthermore, research progress in mitigation and suppression, including system-level thermal management, material-level approach, and structure modification, is discussed. This work aims to support the development of advanced early-warning technologies and to provide guidance for the design of safer next-generation lithium-ion battery systems.

Keywords: lithium-ion batteries; thermal runaway mechanism; thermal runaway prediction; thermal runaway suppression

1. Introduction

The accelerating deterioration of the global environment and climate has intensified the demand for the transition from fossil fuel-based energy systems to renewable energy technologies [1–3]. In this context, lithium-ion batteries have become a cornerstone of modern energy storage [4], serving as the dominant technology for electric vehicles and grid-scale applications owing to their high energy and power densities, long cycle life, negligible self-discharge, absence of memory effects, and low environmental impact [5,6]. However, despite the continued large-scale deployment of lithium-ion batteries, safety concerns have become increasingly critical and are widely recognized as a major barrier to further technological advancement. In particular, safety incidents related to thermal runaway have drawn substantial attention from both academia and industry.

Thermal runaway is an uncontrollable, self-accelerating reaction process in LiBs, characterized by a heat generation rate that far exceeds the heat dissipation capability, leading to a rapid temperature rise. This process is typically accompanied by the release of large quantities of gases [7], which accumulate within the confined battery enclosure and result in continuous pressure buildup. Once the internal pressure exceeds the mechanical tolerance of the battery casing, venting, rupture, or even explosion may occur. Meanwhile, thermal runaway initiates the decomposition of internal battery components, producing a range of flammable and toxic gaseous species. Battery misuse conditions that may trigger a thermal



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runaway are generally classified into three categories: mechanical abuse, electrical abuse, and thermal abuse. Mechanical abuse involves external mechanical forces that damage the battery structure and compromise its physical integrity, such as crushing, puncture, impact, dropping, and bending [8]. Electrical abuse arises from abnormal operating conditions in which voltage or current exceeds design limits, thereby accelerating electrochemical side reactions, including overcharge, over-discharge, high-rate charge/discharge, and external short circuits [9,10]. Thermal abuse occurs when batteries are exposed to abnormal thermal environments, causing the temperature to exceed allowable limits and directly activating exothermic side reactions. When the temperature reaches the decomposition thresholds of key battery materials, a cascade of exothermic reactions is triggered, ultimately leading to thermal runaway [11]. Importantly, these abuse modes are not independent: mechanical damage can induce internal short circuits, thereby initiating electrical and thermal abuse; heat accumulation from electrical abuse may further promote thermal abuse; and excessive temperature rise can degrade battery structures, exacerbating both mechanical and electrical failures. Such coupled interactions form a cascading pathway from initial abuse to thermal runaway [12].

Currently, the safety of LiBs has emerged as a critical bottleneck that must be addressed to enable the large-scale deployment of electric vehicles and grid-scale energy storage systems. Multiple fire incidents triggered by battery thermal runaway [13,14] have highlighted the severe consequences of such failures, underscoring that thermal runaway represents a fundamental constraint on the safe utilization of LiBs. Consequently, elucidating the triggering conditions and underlying mechanisms of thermal runaway, together with developing reliable prediction and effective suppression strategies, is essential for early risk identification and interruption of accident escalation pathways. Although substantial research efforts have been devoted to thermal runaway, several challenges remain. Existing review studies often focus on isolated aspects of thermal runaway, lacking a unified framework that integrates mechanistic insights, early-warning indicators, and mitigation strategies across different spatial and temporal scales.

To ensure the systematic and comprehensive nature of the review, we conducted a rigorous literature search using the Web of Science, Scopus, and IEEE Xplore databases. The search strategy employed a combination of keywords: “lithium-ion battery” AND “thermal runaway” AND (“mechanism” OR “prediction” OR “suppression”). The main time frame was set from 2015 to 2025 to cover the latest developments, while also including the pioneering classic literature. The inclusion criteria prioritized studies involving the mechanism of thermal runaway, multi-sensor fusion detection, data-driven early warning algorithms, as well as thermal management methods, new suppression materials, and advanced battery structures.

The remainder of this paper is organized as follows. Section 2 examines the chain reactions leading from initial failure to thermal runaway. Section 3 reviews current prediction and early-warning approaches, while Section 4 discusses available suppression and mitigation strategies. Section 5 proposes future research directions to address existing gaps, followed by the conclusions in Section 6.

2. Thermal Runaway Mechanism of Lithium-Ion Batteries

Thermal runaway (TR) can be precipitated by a multitude of anomalous conditions, generally categorized into mechanical abuse, electrical abuse, thermal abuse, and aging. While the triggering modes and corresponding mechanisms have been comprehensively elucidated in existing literature, we have synthesized the associated testing standards for these categories in Table 1. However, a comprehensive review of existing abuse methodologies reveals significant discrepancies and inconsistencies among current international

testing protocols. Consequently, there is an imperative need for unified international protocols to standardize critical triggering parameters across various abuse scenarios, such as nail penetration depth and heating rates.

Table 1. Standardized abuse protocols and evaluation metrics.

Category	Test Method	Key Standards	Scenario and Mechanism	Key Refs.
Material Characterization	ARC/DSC	ASTM E1981 [15]; UL 2580 [16]	Adiabatic heating to decouple reaction stages and determine intrinsic T_{onset}/E_a	[17]
Mechanical Abuse	Nail Penetration	SAE J2464 [18]; GB 38031 [19]	Sharp intrusion causing severe internal short circuit and rapid Joule heating	[20]
	Crush/Indentation	ISO 12405 [21]; UL 2580 [16]	Mechanical stress triggers separator failure and localized ISCs.	[22]
	Vibration	UN 38.3 [23]; ISO 16750 [24]	Road fatigue inducing tab tearing or delamination, leading to impedance rise.	[25]
Electrical Abuse	Overcharge	UN 38.3 [23]; IEC 62133 [26]	Induces Li-plating and massive gas generation/swelling prior to thermal runaway.	[27]
	External Short	IEC 62133 [26]; GB 31485 [28]	High discharge current triggers rapid self-heating and separator melting.	[29]
	Over-discharge	GB 31485 [28]; UL 1642 [30]	Cu dissolution forms dendrites, triggering ISC during subsequent recharge.	[31]
Thermal Abuse	Hot Box/Heating	UL 1642 [30]; IEC 62133 [26]	Uniform overheating triggers SEI breakdown and validates thermal limits.	[32]
System Level	Thermal Propagation	GB 38031 [19]; GTR 20 [33]	Cell-to-pack spread via heat conduction/convection; tests thermal barrier efficacy.	[34,35]

Notwithstanding the prevailing variations in international abuse testing protocols, the ultimate consequence of these external triggers remains consistent: the disruption of the battery’s internal thermodynamic stability. Therefore, deciphering how macroscopic external stimuli induce internal electrochemical destabilization at the microscopic level is pivotal for comprehending the fundamental nature of thermal runaway. The mechanism of TR in lithium-ion batteries involves complex thermal equilibria and chain reactions [36]. Accordingly, this section will provide a detailed elaboration on the mechanisms driving thermal runaway under abuse conditions.

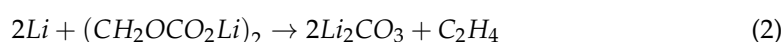
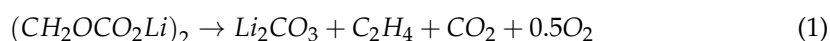
2.1. Capacity Degradation

Elevated temperatures accelerate lithium-ion transport between the cathode and anode, leading to enhanced self-discharge and accelerated capacity degradation. Existing studies indicate that irreversible active lithium loss and structural degradation of electrode materials jointly drive sustained capacity fade, typically accompanied by increased interfacial impedance and overall internal resistance [37]. High temperatures also intensify electrolyte decomposition and SEI evolution, promoting inhomogeneous side reactions and

interfacial layer thickening on both electrodes, which has been shown to correlate directly with capacity decay [38–40]. Overall, high-temperature-induced capacity degradation arises from the coupled effects of lithium loss, active material degradation, and interfacial impedance growth.

2.2. Decomposition of Solid Electrolyte Interface (SEI)

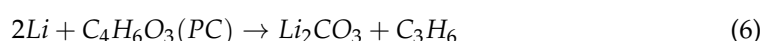
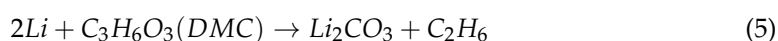
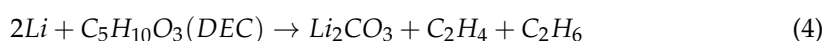
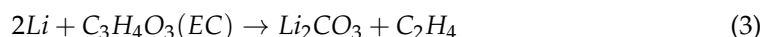
The materials inside the lithium-ion battery begin to decompose and generate heat as the temperature further increases. The first exothermic reaction is the decomposition of the SEI film, and its metastable components will decompose and release heat at 57 °C [41]. The reaction formula is introduced below:



Decomposition reaction of the SEI membrane detects significant heat release at 80 °C and reaches a peak heat production at 100 °C [42]. As the SEI film decomposes, the available lithium at the negative electrode of the battery comes into contact with the electrolyte and regenerates an irregular and unstable new SEI film [43]. This decomposition and regeneration reaction will continue to circulate at high temperatures until the battery experiences thermal runaway [44].

2.3. Negative Electrode-Electrolyte Reaction

Oka et al. [45] reported that overcharging markedly intensifies both oxidative and reductive electrolyte decomposition, thereby inducing lithium metal reactions and additional electrode corrosion. Moreover, decomposition of the solid electrolyte interphase (SEI) exposes the negative electrode active material, which subsequently reacts with embedded lithium in the presence of the electrolyte, further accelerating interfacial degradation [46]. The chemical equation is described by



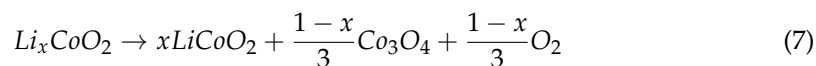
2.4. Separator

Polyethylene (PE) and polypropylene (PP) are widely used as commercial battery separators, with melting points of approximately 130 °C and 170 °C, respectively [47]. Separator melting is an endothermic process that can temporarily slow down, or even reverse, the battery temperature rise [48]. However, as the temperature continues to increase, separator shrinkage may occur, leading to direct contact between the positive and negative electrodes and triggering internal short circuits. This process re-accelerates heat generation and becomes a critical heat source driving uncontrolled contact-induced heating and thermal runaway [49].

2.5. Decomposition of Positive Electrode

Cathode material chemistry plays a critical role in determining the thermal safety performance of lithium-ion batteries, thereby influencing both the likelihood and severity of thermal runaway. Doughty et al. [50] systematically compared the thermal safety characteristics of batteries employing different cathode materials.

Lithium cobalt oxide (LCO) was the earliest commercially deployed cathode material for lithium-ion batteries. However, owing to its poor thermal stability—particularly under high-temperature or overcharged conditions—it is no longer widely adopted in large-capacity power battery applications. The thermal decomposition reaction of lithium cobalt oxide can be expressed as:



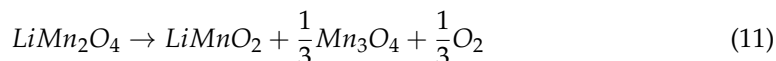
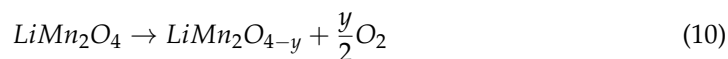
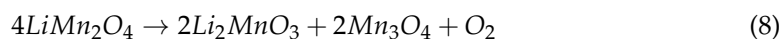
During the thermal decomposition of lithium cobalt oxide (LCO) cathodes, Co^{4+} is reduced to Co^{3+} , accompanied by the release of oxygen and a substantial amount of heat. In comparison, Nickel–Cobalt–Manganese (NCM) cathode materials exhibit relatively lower thermal decomposition intensity than LCO. Among the transition metal species, Ni ions are more electrochemically active than Co and Mn ions and are therefore more susceptible to valence-state changes under thermal abuse conditions. As a result, the dominant reactions between NCM cathodes and the electrolyte are associated with the reduction of Ni^{4+} to lower valence states, such as Ni^{2+} . Noh et al. [51] systematically investigated the influence of Ni content on the thermal stability of $\text{Li}(\text{Ni}_x\text{Co}_y\text{Mn}_z)\text{O}_2$, with the results summarized in Table 2.

Table 2. Decomposition enthalpy (ΔH) and the initial decomposition temperature (T_{onset}) of $\text{Li}(\text{Ni}_x\text{Co}_y\text{Mn}_z)\text{O}_2$ with different compositions [51].

Material	ΔH (J·g ^{−1})	T_{onset} (°C)
$\text{Li}_{0.37}[\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}]\text{O}_2$	512.5	300
$\text{Li}_{0.34}[\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}]\text{O}_2$	605.7	285
$\text{Li}_{0.30}[\text{Ni}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}]\text{O}_2$	721.4	251
$\text{Li}_{0.26}[\text{Ni}_{0.7}\text{Co}_{0.15}\text{Mn}_{0.15}]\text{O}_2$	826.3	230
$\text{Li}_{0.23}[\text{Ni}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}]\text{O}_2$	904.8	220
$\text{Li}_{0.21}[\text{Ni}_{0.85}\text{Co}_{0.075}\text{Mn}_{0.075}]\text{O}_2$	971.5	215

Kim et al. [52] investigated the thermal stability of $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$ using differential scanning calorimetry (DSC) and identified three distinct exothermic peaks at approximately 325 °C, 362 °C, and 458 °C. The first two exothermic peaks were attributed to the thermal decomposition of the NCM cathode material.

Thermal decomposition of lithium manganese oxide (LMO) cathodes involves the reduction of Mn^{4+} accompanied by oxygen release, and the corresponding reaction equations can be expressed as follows:

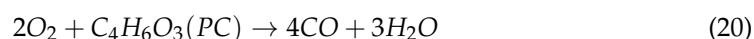
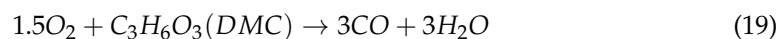
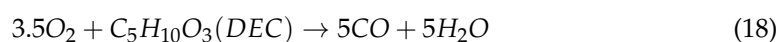
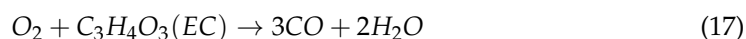
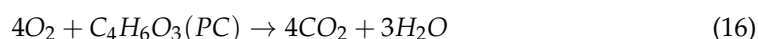
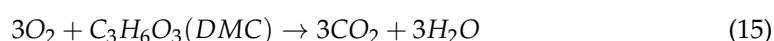
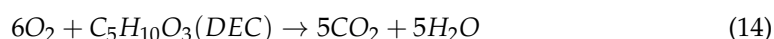
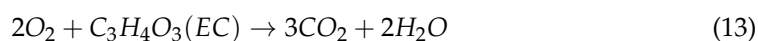


By using high-concentration electrolytes or adding functional components to optimize the chemical environment of the electrolyte, it is an effective way to enhance the thermal stability and cycle life of LMO [53]. Finegan et al. [54] further quantified the thermal decomposition behavior of LMO, reporting a maximum heat release of up to 450 J·g^{−1} over a temperature range of 150–300 °C.

In contrast to the cathode materials discussed previously, LFP exhibits superior thermal stability attributed to the strong covalent P-O bonds within its olivine crystal structure. Its thermal decomposition onset temperature is significantly higher, ranging from 500 to 800 °C. Furthermore, LFP is characterized by a relatively low decomposition enthalpy, indicating minimal heat release during the reaction. Even following irreversible phase transitions at temperatures exceeding 500 °C, the robust P-O framework remains intact and prevents structural collapse [55].

2.6. Decomposition Reaction of Electrolyte

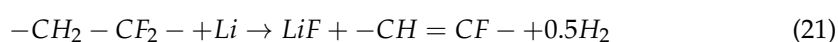
The electrolyte serves as the ionic conduction medium in lithium-ion batteries and typically consists of lithium salts dissolved in organic carbonate solvents. Thermal decomposition of the electrolyte can release substantial amounts of heat and gaseous products, thereby contributing significantly to thermal runaway [56]. The following reactions describe the complete oxidation of electrolyte components to CO₂ as well as incomplete oxidation pathways leading to CO formation, where EC denotes ethylene carbonate, DEC denotes diethyl carbonate, DMC denotes dimethyl carbonate, and PC denotes propylene carbonate.



The decomposition mechanism is highly sensitive to the electrolyte formulation. Equations (13)–(20) depict pathways typical for standard LiPF₆-carbonate systems, where LiPF₆ dissociation yields Lewis acidic PF₅ that catalyzes solvent decomposition [57]. In contrast, alternative salts such as LiFSI or varying solvent structures like linear versus cyclic carbonates exhibit distinct thermal stabilities and gas evolution profiles. Thus, actual decomposition products depend heavily on the specific salt–solvent–additive matrix. Based on Lewis theory, the PF₅ generated by the decomposition of LiPF₆ attacks the oxygen in the C-O bond, thereby accelerating the decomposition of the electrolyte. Botte et al. [58] further showed via DSC that LiPF₆ undergoes an initial endothermic process prior to exothermic decomposition in EC: EMC electrolytes, whereas in EC:DEC:EMC (1:1:3), exothermic reactions initiate at around 256 °C with a total heat release of 210 J·g^{−1}.

2.7. Decomposition Reaction of Adhesive

Polyvinylidene fluoride binders can enhance the structural integrity and thermal stability of battery electrodes at elevated temperatures, thereby improving the overall mechanical robustness of lithium-ion batteries. When the temperature exceeds 260 °C, intercalated lithium in negative electrode undergoes exothermic reactions accompanied by hydrogen gas evolution [59,60]:





2.8. Electrolyte Combustion

Upon melting and rupture of the aluminum current collector, high-temperature and high-pressure gases may be violently ejected from the battery, entraining cathode active materials and forming visible black smoke [61]. At the battery-pack level, combustion behavior is generally characterized by multiple stages, including cell swelling, ignition, stable combustion, and subsequent attenuation or extinction [62]. Meng et al. [63] investigated the combustion behavior of lithium iron phosphate (LFP) batteries using an indoor fire test platform and reported that fully charged cells (100% SOC) underwent thermal runaway at approximately 166.8 °C. This event was accompanied by the release of large quantities of flammable gases and electrolyte, leading to jet ignition above the safety valve with a peak flame temperature of 943.1 °C and a maximum heat release rate (HRR) of 12.3 kW. In addition, Hu et al. [64] demonstrated that gas generation during thermal runaway is highly dependent on abuse conditions, temperature, electrode materials, and electrolyte composition, highlighting the strong coupling between battery chemistry, operating conditions, and combustion behavior.

2.9. Kinetic Analysis

Thermal runaway progression follows Arrhenius kinetics, where reaction rates depend exponentially on temperature. It is crucial to recognize that the degradation pathways described in Sections 2.2–2.6 are not static; they are dynamically governed by these kinetic principles. As the temperature rises, the dominant reaction mechanism shifts—for instance, transitioning from mild SEI decomposition to violent electrolyte oxidation—thereby altering the specific reaction products and heat generation rates. This process is categorized into three stages based on activation energy (E_a). Stage I is SEI decomposition with a relatively low $E_a \approx 50\text{--}70 \text{ kJ}\cdot\text{mol}^{-1}$, acting as the initiating event, and Stage II involves the anode–electrolyte reaction ($E_a \approx 80\text{--}120 \text{ kJ}\cdot\text{mol}^{-1}$) [65], driving the system into self-sustained heating. Stage III is cathode decomposition, where high-nickel NCM ($E_a \approx 150 \sim 250 \text{ kJ}\cdot\text{mol}^{-1}$) demonstrates significantly lower stability compared to LFP [66], leading to rapid thermal destabilization.

These kinetic parameters are quantitatively derived from Accelerating Rate Calorimetry (ARC) and differential scanning calorimetry (DSC) [67,68]. By isolating heat flow peaks under adiabatic or constant-heating conditions, researchers can decouple overlapping reactions. The specific E_a and pre-exponential factor A for each component is then determined through linear regression of $\ln(dT/dt)$ versus $1/T$, establishing characteristic kinetic profiles for different cell chemistries.

Integrating kinetic parameters into the battery management system enables predictive thermal safety. Specifically, the pre-exponential factor A and E_a feed into the chemical heat generation term of electro-thermal models. By solving governing equations in real time, the BMS distinguishes irreversible decomposition heat from reversible Joule heating and extrapolates reaction trends to predict the Time-to-Thermal-Runaway (TTR) [69]. Furthermore, kinetic analysis establishes dynamic safety boundaries: if the temperature rise rate exceeds a threshold derived from the Stage II activation energy, it signals the dominance of self-accelerating reactions and triggers an early warning before catastrophic temperatures are reached [70].

3. Prediction of Thermal Runaway

3.1. Characteristic Signal-Based Method

3.1.1. Voltage Signal

During the early stage of battery failure, the terminal voltage typically remains stable or may even exhibit a slight increase, as accelerated interfacial reactions between the electrolyte and electrode materials partially compensate for voltage decline [71,72]. With continued temperature rise, intensified side reactions disrupt the electrochemical equilibrium and may induce internal short circuits, resulting in a pronounced voltage drop [73]. Based on these voltage evolution characteristics, several voltage-based early-warning methods have been proposed. Du et al. [74] developed a Gaussian smoothing filter (GSF) for voltage signal denoising, followed by a feature exponential function (FEF) to enhance subtle fault-related voltage signatures, and employed dynamic time warping (DTW) for automatic fault identification and localization. Yu et al. [75,76] proposed a thermal runaway warning approach based on battery relaxation voltage analysis to estimate short-circuit resistance, which was validated across multiple battery brands and demonstrated robustness under varying temperature and current conditions.

Voltage signals face high-frequency electromagnetic interference from motor inverters, requiring robust filtering to prevent false alarms. Additionally, voltage sensors are inherently insensitive to early micro-short circuits, necessitating fusion with other signals for timely fault detection.

3.1.2. Temperature Signal

Gulsoy et al. [77] monitored both internal and external battery temperatures using thermocouples embedded in cylindrical cells. Compared with thermocouples, fiber optic sensors offer advantages in size, flexibility, and environmental robustness, making them well-suited for battery temperature monitoring. Consequently, various fiber optic sensing technologies, such as fiber Bragg gratings (FBGs), Fabry–Perot fibers, and fluorescent fibers, have been explored for lithium-ion battery applications. Mei et al. [78] fabricated compact FBG-based sensors using femtosecond laser processing, while Li et al. [79] developed a fluorescence-based fiber optic sensor employing frequency-upconverting nanoparticles to directly measure internal battery temperature. Direct measurement of internal temperature is of critical importance for early thermal runaway warning. However, the practical deployment of implanted fiber optic sensors remains constrained by electrolyte corrosion and relatively high system cost.

Surface sensors suffer from thermal lag, failing to reflect core temperature in real-time. Optimal placement requires positioning sensors near electrode tabs or integrating them into cooling channels, though this significantly increases assembly complexity.

3.1.3. Mechanical Signal

Current approaches for capturing mechanical signals in lithium-ion batteries mainly include measuring the expansion force of prismatic cells using dedicated fixtures and detecting strain via fiber gratings or resistance strain gauges [80], as illustrated in Figure 1. Sun et al. [81] demonstrated that internal pressure peaks consistently precede temperature peaks during failure evolution. Jin et al. [82] analyzed the expansion force characteristics of pouch-cell modules in the early stage of thermal runaway and showed that mechanical signals exhibit superior noise immunity. Choi et al. [83] developed a numerical model to simultaneously characterize battery lid deformation and internal pressure evolution during thermal runaway propagation. Li et al. [84] achieved accurate monitoring of internal temperature and pressure through embedded sensors, providing valuable data for early warning applications. In addition, Lin et al. [85] revealed that the superposition

of expansion forces from adjacent cells is the primary cause of bimodal expansion force behavior during heat propagation in battery packs. Although mechanical sensing shows clear potential, sensor integration remains technically challenging [86]. Overall, these studies indicate that mechanical deformation is a sensitive indicator of internal battery states. Compared with temperature and voltage signals, mechanical responses typically emerge earlier; however, the exclusive use of mechanical signals for thermal runaway early warning is still limited in practice due to robustness issues and deployment constraints [87–89]. Integration requires pre-loaded sensors within rigid modules, which may compromise pack energy density. A major algorithmic challenge is distinguishing fault-induced irreversible expansion from the normal reversible swelling of cells during cycling.

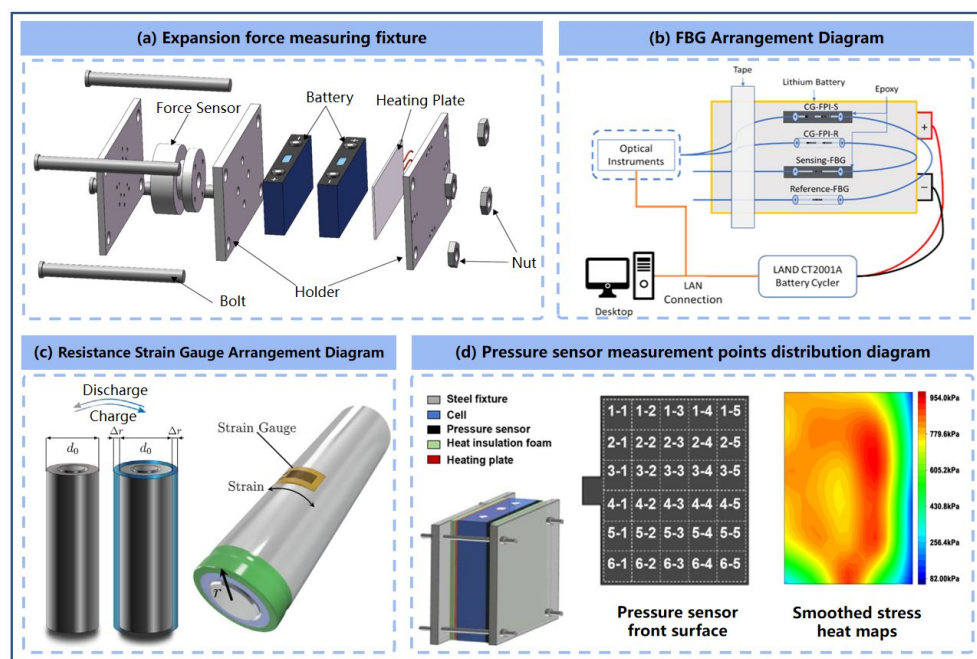


Figure 1. Methods for acquiring mechanical signals during battery swelling. (a) Expansion force measuring fixture [85]; (b) FBG arrangement diagram [86]; (c) Resistance strain gauge arrangement diagram [87]; (d) Pressure sensor measurement points distribution diagram and smoothed stress heat maps [88].

3.1.4. Gas Signal

Commonly used gas sensors for LiB monitoring include semiconductor, optical, and electrochemical sensors [90]. Gas-based early-warning capability has been widely demonstrated. Yang et al. [91] reported early detection of H_2 during battery overcharge, followed by CO release approximately 579 s before thermal runaway, confirming the feasibility of gas-signal-based early warning. Zhang et al. [92] proposed a multi-level warning strategy by coupling H_2 and CO detection with electrical and thermal indicators. In addition to these gases, CO_2 has been identified as another major component released during thermal runaway [93]. Fernandes et al. [94] further revealed a multi-stage gas release behavior in overcharged lithium iron phosphate batteries, where the initial venting was dominated by dimethyl carbonate (DMC) and ethyl methyl carbonate (EMC), followed by H_2 , CO_2 , C_2H_4 , and CO. Furthermore, Li et al. [95] systematically summarized the evolution of gas components associated with side reactions in standard $LiPF_6$ -carbonate electrolytes across different temperature stages, as illustrated in Figure 2. The specific gas composition and onset temperatures are strongly dependent on the electrolyte chemistry. Thus, gas-based early warning thresholds must be calibrated according to the specific cell chemistry.

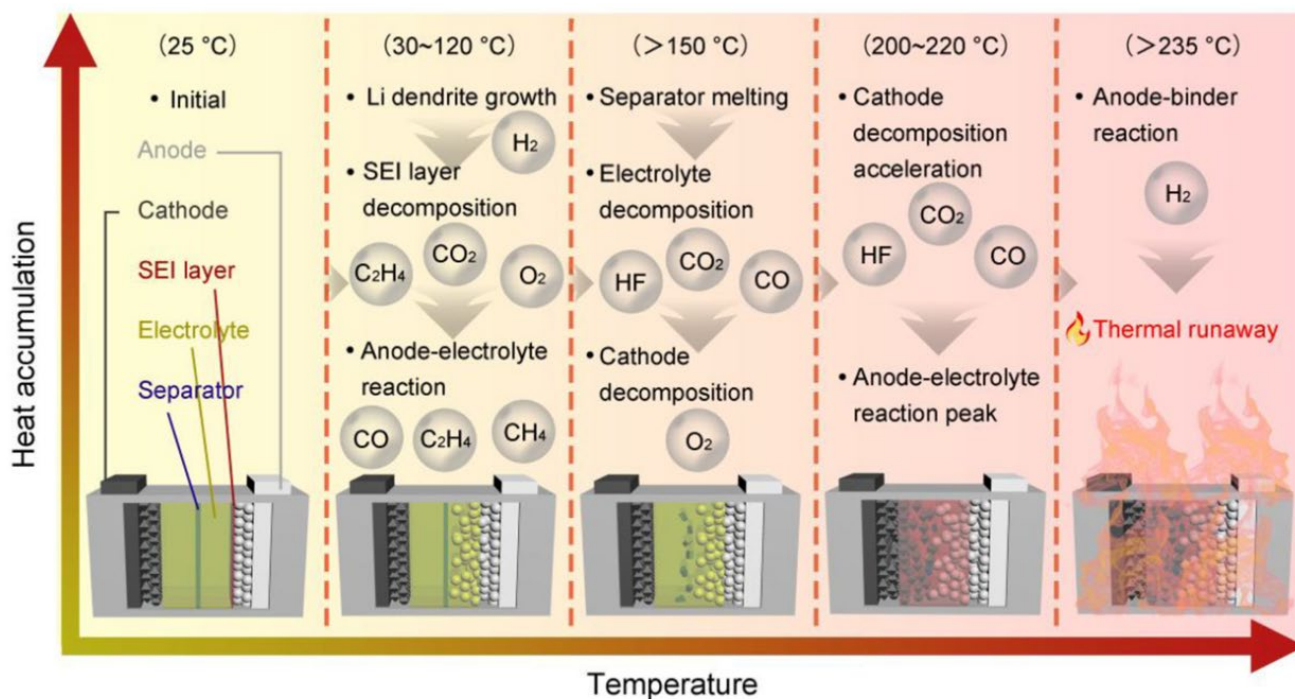


Figure 2. Gas composition from side reactions of PF in standard Li_6 -carbonate electrolytes of batteries at different temperature nodes [95].

Monitoring gases released during battery thermal runaway provides an effective early-warning approach, as gas species and their concentration evolution serve as sensitive indicators of internal battery states [96]. Accordingly, gas sensing plays a critical role in enabling timely responses by battery management systems under hazardous conditions. Song et al. [97] analyzed gas generation characteristics under various abuse scenarios and proposed an optimized gas sensor selection strategy tailored to batteries with different material chemistries, thereby improving early-warning accuracy. Liao et al. [98] identified C_2H_4 , CH_4 , and CO as characteristic gases and evaluated the corresponding warning lead times for lithium-ion battery thermal runaway. Despite these advances [90], gas-based early warning inherently relies on the accumulation of gaseous species to detectable concentration levels. In large-scale energy storage stations, gas dilution and dispersion may delay sensor response, thereby compromising early-warning effectiveness. Sensor deployment strategies and spatial configuration remain critical yet challenging issues. Gas detection is primarily limited by diffusion delay, requiring sensors to be placed in exhaust paths or near vents. Furthermore, cross-sensitivity to volatile organic compounds (VOCs) from sealing materials necessitates high-selectivity sensors.

3.1.5. Acoustic Signal

Non-contact acoustic sensors are employed to capture characteristic acoustic emissions during failure evolution, as illustrated in Figure 3. Zhu et al. [99] analyzed the acoustic signatures of prismatic battery thermal runaway using high-frequency wavelet packet energy ratio analysis. Su et al. [100] proposed a rapid and cost-effective acoustic-based warning method for megawatt-scale battery systems, achieving effective thermal runaway detection with an accuracy of 92.31%. Similarly, an acoustic early warning framework reported detection accuracies exceeding 90%, demonstrating the feasibility of sound-based approaches for timely fault identification [101].

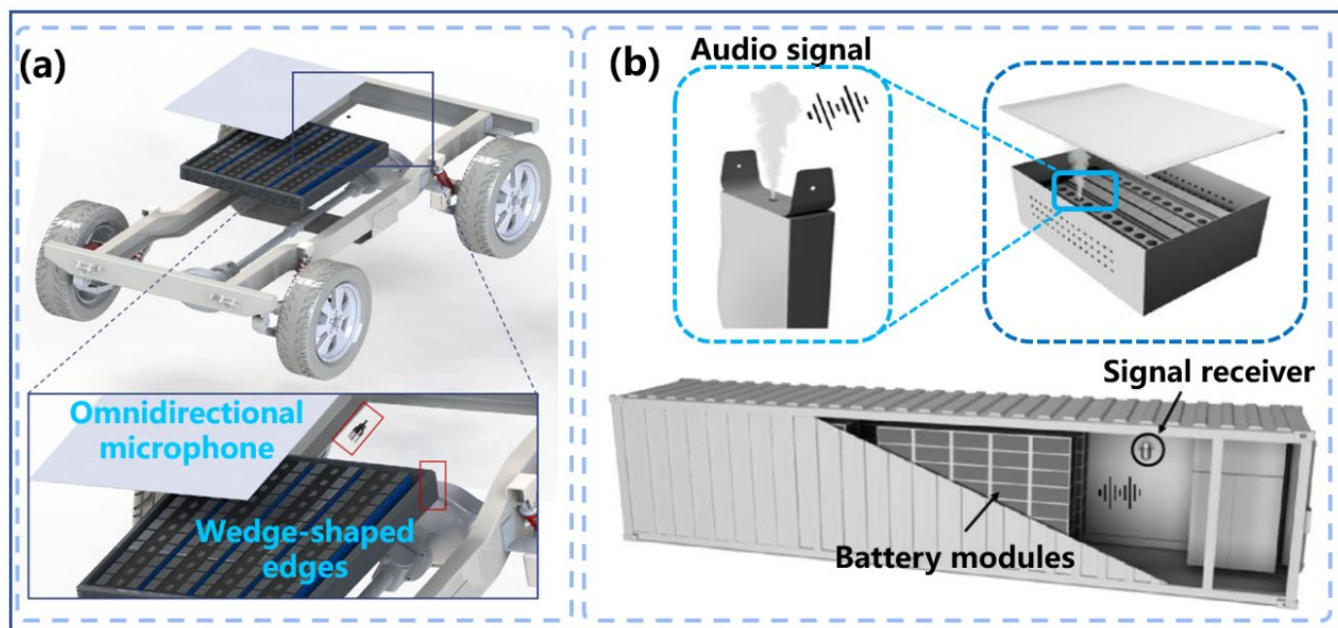


Figure 3. Placement methods of acoustic sensors. (a) In new energy vehicles [99]. (b) In energy storage power stations [100].

Beyond detection, acoustic sensing has also been explored for fault localization. Lyu et al. [102] developed an acoustic-based alarm and localization method by deploying multiple sensors within an energy storage compartment to capture pressure-release sounds, achieving a maximum localization error of approximately 0.1 m. A two-stage acoustic early warning strategy was proposed to further improve detection reliability [103]. Similar with the gas detection, practical deployment of sound-based early warning remains challenging. Acoustic sensors are inherently susceptible to background noise and interference in complex operational environments, which can degrade detection accuracy. Moreover, the limited availability of real-world thermal runaway event data restricts model training, potentially constraining the robustness and generalization capability of machine learning-based approaches.

Acoustic sensors face severe background noise in EVs. Advanced spectral filtering is required, making this technology currently more viable for stationary ESS rather than dynamic automotive applications.

3.1.6. Optical Signal

Machine vision has been applied to battery surface defect detection [104,105], and image-, video-, and thermal-imaging-based early warning has become an important direction [106]. These methods capture morphological and thermal abnormalities to enable early detection. Chen et al. [107] analyzed thermal imaging differences under varying SOC conditions, as shown in Figure 4a. Hong et al. [108] developed a YOLOv5-based algorithm that identifies normal, swollen, and thermal-runaway states with over 98% accuracy, as shown in Figure 4b. Ding et al. [109] incorporated thermal imaging into a multimodal neural network, integrating it with temperature and voltage data to achieve multi-step thermal-runaway prediction.

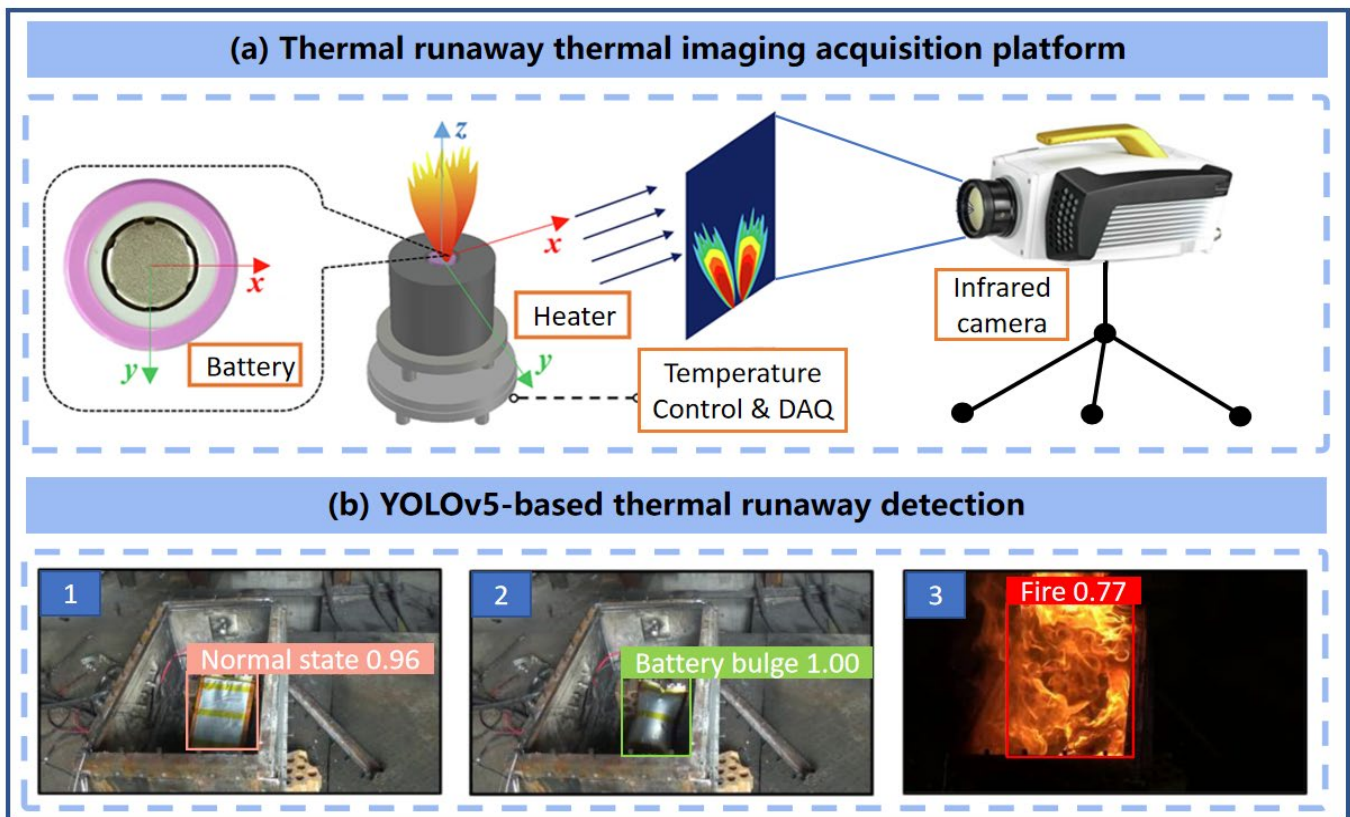


Figure 4. Optical signals-based thermal runaway early warning. (a) Schematic of the thermal runaway thermal imaging acquisition platform for lithium-ion batteries [107]; (b) YOLOv5-based detection results: 1—normal state (0.96), 2—battery bulge (1.00), 3—fire (0.77) [108].

Optical methods are strictly limited by line-of-sight requirements. Densely packed modules obstruct internal views, rendering this method suitable primarily for open-rack storage systems or lab testing rather than enclosed EV battery packs.

3.1.7. Multidimensional Information Fusion

While Sections 3.1.1–3.1.6 have extensively explored individual sensing methodologies, relying exclusively on a single signal source presents inherent risks for practical battery management systems. As systematically compared in Table 3, each detection principle possesses intrinsic limitations that can compromise safety reliability. Traditional voltage and temperature monitoring, despite their maturity and low cost, are often constrained by insensitivity to micro-shorts and significant thermal hysteresis, leading to delayed responses. Conversely, advanced methodologies such as gas analysis and mechanical expansion sensing offer superior lead times (5–15 min and 2–10 min, respectively) but are plagued by challenges like sensor breakage, integration complexity, and high costs. Similarly, acoustic and optical techniques, while enabling non-contact detection, are highly susceptible to environmental interference such as background noise and line-of-sight obstructions.

Table 3. Comparative analysis of characteristic metrics and application scenarios for single-dimension sensing methodologies.

Methodology	Principle	Key Advantages	Key Disadvantages	Typical Lead Time	Cost	Recommended Application Scenario
Voltage	Terminal voltage drop	No extra sensors; High maturity	Insensitive to micro-shorts	Short (<1 min)	Low	All Scenarios
Temperature	Surface/Tab heat generation	Direct indicator; Simple	Thermal lag; Surface $\neq$ Core	Short (<1 min)	Low	All Scenarios
Gas	Venting gas detection	Earliest warning capability	Sensor poisoning; Diffusion delay	Long (5–15 min)	Medium	Large-scale ESS; Enclosed battery rooms
Mechanical	Cell expansion/Strain	High sensitivity to gas	Hysteresis; Hard to integrate	Medium (2–10 min)	High	Lab Testing; High-end EVs with rigid frames
Acoustic	Venting sound emission	Non-contact; Fault localization	Background noise interference	Medium	Medium	Stationary ESS; Closed compartments
Optical	Visual/IR imaging	Intuitive location	Line-of-sight required	Varies	High	Lab Testing; Open-rack storage systems

Thermal runaway is a multi-physics coupled process; thus, relying on any single feature risks false alarms or missed events under sensor faults or environmental interference [110,111]. Recent studies have established hierarchical early-warning frameworks based on multi-signal fusion, reporting high detection accuracy and enabling adaptive safety warnings through quantitative threshold design [112,113]. For instance, Feng et al. [114] integrated voltage, temperature, combustible gas, and pressure for diagnostic modeling, while Jia et al. [115] introduced a composite fiber Bragg grating parameter enabling simultaneous temperature and strain measurement. Li et al. [116] showed that mass change detection can issue warnings roughly two minutes sooner than temperature-based alerts. Xu et al. [117] constructed long-term consistency metrics for battery packs using voltage, temperature, internal resistance, and capacity, applying fuzzy AHP and entropy weighting to identify the lowest-consistency cell and thereby provide early thermal-runaway warnings.

3.2. Model-Based Methods

Model-based methods focus on identifying early abnormal battery states using equivalent circuit, electrochemical, and electrothermal coupling models [118]Equivalent circuit models are widely applied due to their low complexity and suitability for online estimation and fault detection [114,119], as shown in Figure 5a. Xu et al. [120] employed multiscale techniques by combining a fast PI observer with long-term SOC accumulation to enhance fault diagnosis. Recent work has also shifted from time-domain to frequency-domain analysis [121]. Electrochemical impedance spectroscopy (EIS), which captures intrinsic electrochemical changes, offers new opportunities for thermal-runaway warning, as shown in Figure 5b. Srinivasan et al. [122] showed that phase shifts at 40 Hz decrease with rising temperature due to reduced graphite–anode impedance. Spinner et al. [123] further demonstrated that the imaginary part of impedance decreases as internal temperature increases, enabling warnings before the onset of self-heating under abuse conditions.

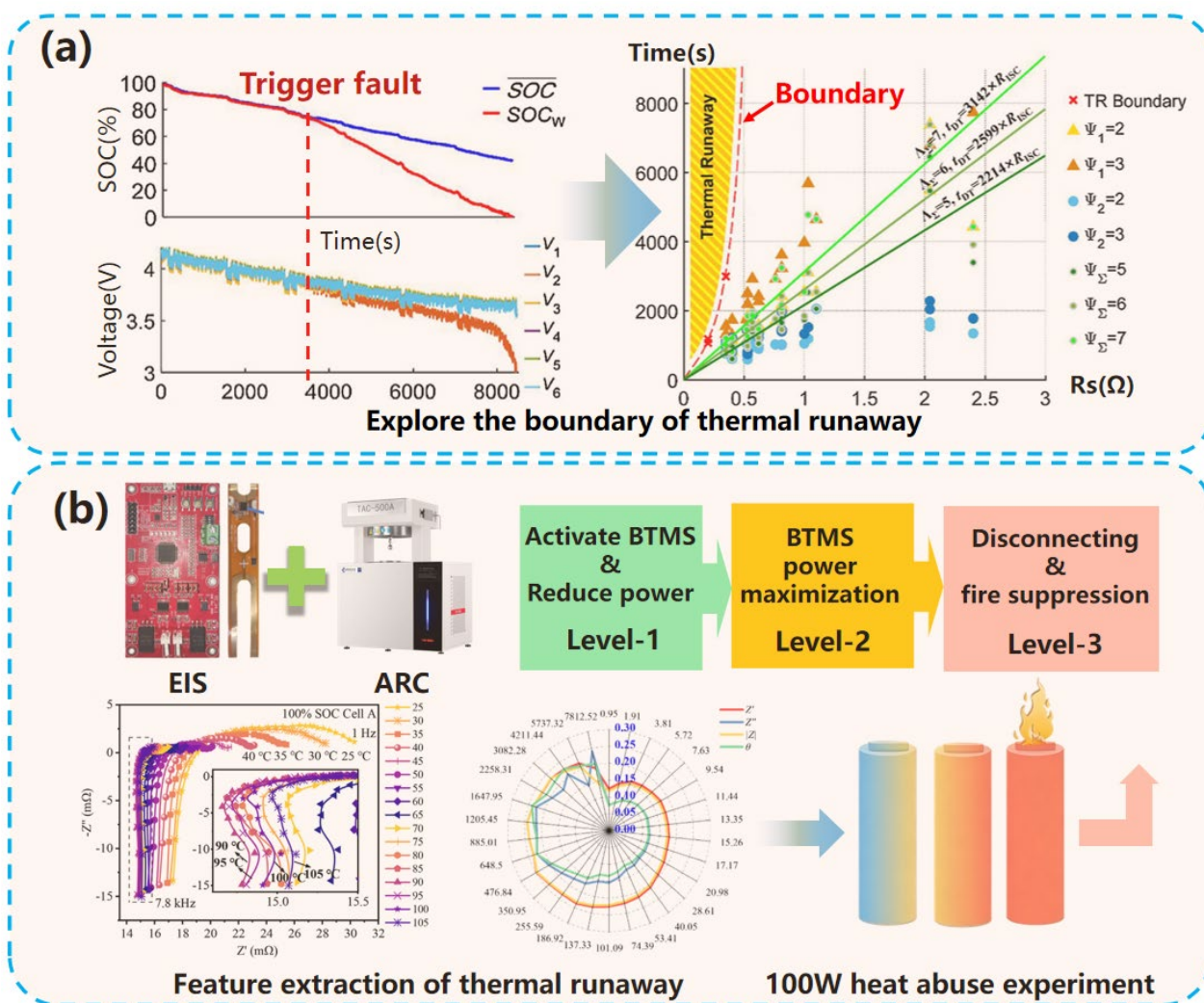


Figure 5. Model-based method. (a) thermal runaway prediction based on equivalent circuit models [114]; (b) EIS-based early warning method [121].

Electrothermal coupling models based on equivalent circuits enable early anomaly detection before thermal runaway [124–126]. Chen et al. [127] constructed a 3D electrothermal model to simulate temperature fields under different charge/discharge rates and assess thermal propagation from a failed cell to neighboring cells. Zeng et al. [128] developed a 3D electrochemical–thermal model capturing the transition from normal charging to early overcharge, showing that irreversible heat dominates during normal operation, whereas Mn dissolution and Li deposition become primary heat sources during early overcharge, supporting a multi-stage overcharge warning strategy. Ren et al. [129] identified kinetic parameters for exothermic reactions using the Kissinger method and nonlinear fitting, establishing a prediction model that accurately forecasts critical thermal runaway temperatures. Feng et al. [130] proposed a bidirectionally coupled electrochemical–thermal failure prediction model capable of forecasting voltage drops, abnormal temperature rises, and quantifying heat contributions from internal short circuits during thermal runaway.

3.3. Data-Driven Thermal Runaway Prediction Method

3.3.1. Analysis-Based Method

To meet the power and capacity requirements of power batteries, cells are typically combined in parallel or series configurations. However, due to manufacturing inconsistencies [131] and temperature variations during operation [132], voltage inconsistencies

are amplified. Consequently, analyzing battery pack voltage data streams is crucial for predicting thermal runaway in electric vehicles [133]. Wang et al. [134] proposed a voltage anomaly detection method based on modified sliding-window Shannon entropy, employing the z-score method to determine anomaly coefficient thresholds for early warning of potentially thermally runaway batteries. Similarly, Tang et al. [135] proposed a battery thermal runaway early warning strategy based on improved information entropy weighting. This approach identifies abnormal battery cells by averaging longitudinal outliers and evaluates pack consistency by integrating multidimensional operational data to reduce false positives. Its effectiveness was ultimately validated through real-vehicle thermal runaway experiments. Correlation coefficients quantify linear relationships among battery operating parameters. Li et al. [136] employed a multi-fault online diagnosis method combining non-redundant measurement topology with weighted Pearson correlation coefficients (WPCC). By weighting measurement data with varying forgetting factors, they localized distinct circuit faults. Although information analysis methods offer advantages such as simplicity, computational efficiency, and independence from complex electrochemical models, they heavily rely on large amounts of high-quality historical data and assume specific statistical distributions.

3.3.2. Machine Learning-Based Methods

With increasing data volumes and battery system complexity, machine learning has become a major direction for thermal runaway early warning. Its strengths in nonlinear modeling, automatic feature extraction, and efficient time series analysis enable effective real-time monitoring [137]. This section reviews deep learning, hybrid neural networks, time series models, few-shot and transfer learning, and physics-informed approaches.

Deep learning can automatically extract features from raw data, making it suitable for modeling the complex nonlinear evolution of thermal runaway. Ma et al. [138] combined wavelet analysis with an attention-based deep learning model to map time frequency representations for voltage and temperature joint alarms, achieving 8 to 13 min of advance prediction in practice. Building on architectural complementarity, Huang et al. [139] integrated convolutional neural networks, Transformer structures, and multilayer perceptrons to predict the remaining time before thermal runaway under high-rate conditions. Time series models naturally capture temporal dependencies in voltage and temperature signals. Zheng et al. [140] employed convolutional LSTMs with sliding windows to develop a charging protection strategy for hybrid vehicles, showing strong robustness to noise and enabling threshold design for early thermal-runaway prediction through residual analysis. However, training such models requires substantial computation and large labeled datasets, which is difficult given the scarcity of real thermal runaway incidents and increases the risk of overfitting.

Few-shot learning and transfer learning offer solutions to the scarcity of labeled thermal runaway data by enabling models to learn effectively from limited samples. Dong et al. [141] proposed an MDTL-FSL framework that transfers knowledge from multiple related source domains, selects source domains with distributions close to the target to avoid negative transfer, and uses adversarial learning and meta-learning to extract domain-invariant features and optimize decision boundaries. This method achieves early warnings across diverse battery types and conditions. Yang et al. [142] introduced a data augmentation strategy combining transfer learning with conditional GANs, where transfer learning mitigates target-domain data scarcity and GANs generate physically plausible fault samples. This approach improves recall and F1 scores under few-shot conditions, though excessive reliance on sparse samples may reduce discriminative ability on abundant

normal data, and transfer learning lacks interpretability and may degrade performance when domain gaps are large.

Physics-informed approaches improve interpretability by incorporating physical equations into models. Luo et al. [143] embedded physical relationships among strain, temperature, and state of charge into the loss function, enhancing accuracy and generalization of a multi-physics state model. Residuals from this high-precision PINN enabled reliable early warnings of overcharge-induced thermal runaway. Zhang et al. [144] coupled an equivalent circuit model with a thermal model and extended a Kalman filter observer with a bidirectional LSTM to learn uncertainties, allowing the system to distinguish faults from disturbances. Although physics-informed fusion improves extrapolation and interpretability, it depends on accurate physical models.

Although the accuracy of machine learning models is high in laboratory settings, their practical application in battery management systems (BMS) still faces numerous challenges. One of the main obstacles is the domain bias caused by battery aging or differences in battery types, which requires the use of strategies such as transfer learning to maintain their performance throughout their entire lifecycle [145]. Additionally, environmental noise may mimic fault characteristics, which necessitates the use of robust algorithms with signal denoising capabilities and trained with synthetic noise [146]. Furthermore, given the severe class imbalance problem in rare thermal runaway events, evaluation metrics must prioritize F1 score over simple accuracy to ensure safety while also considering user experience [147]. Before industrial deployment, standardized tests under various operating conditions are required to verify the model's immunity to non-fault anomalies [148].

3.3.3. Large Model-Driven Method

Compared with traditional threshold-based and single-sensor thermal runaway warning methods, model-driven systems built on large-scale battery models achieve a substantial advance. Notably, the scale of large models and their dataset requirements differ substantially from those of conventional machine learning models. Traditional machine learning models typically involve parameter scales below 10^6 , whereas recent Transformer-based architectures designed for battery applications generally range from several million to tens of millions of parameters, as reported in recent studies. Correspondingly, these models often rely on larger datasets on the order of 10^5 to 10^6 training samples or long-duration electrochemical time-series data to ensure stable convergence and robust cross-condition performance. Conventional approaches cannot effectively capture early and subtle fault features under multi-factor coupling, leading to delayed detection and high false alarm rates [149,150]. In contrast, large-scale models use massive parameters to approximate complex nonlinear behavior and extract microscopic indicators of failures such as consistency degradation, internal short circuits, and overcharge from multi-source data. This enables a shift from passive threshold triggering to proactive mechanism-based inference, greatly improving both the accuracy and early-warning capability of thermal runaway prediction [151,152].

The core of this advanced system is the construction of a high-quality big data foundation. On cloud platforms, multi-source heterogeneous data from real vehicle operation and battery simulations are integrated and processed to form a large-scale dedicated battery dataset [153,154]. This dataset spans the full lifecycle from production to recycling and captures battery behavior under both typical and extreme conditions [155], providing essential support for training large models capable of understanding complex battery mechanisms. The increased architectural complexity of large models also results in higher inference latency. Reported inference times on embedded or edge computing platforms commonly range from several tens to over one hundred milliseconds, depending on model depth and

input sequence length. In contrast, conventional methods such as Support Vector Machines (SVMs) and Random Forest (RF) typically achieve inference latency below 10 ms under similar deployment conditions. Consequently, while traditional machine learning models can be deployed directly on local battery management system microcontrollers, large model-driven approaches often require cloud-edge collaborative architectures to balance computational demand and deployment feasibility. This factor represents an important consideration for practical implementation.

At the large model layer, the cloud uses this high-quality dataset to drive continuous pre-trained model optimization [156]. To meet industrial needs, domain-specific fine-tuning based on general foundational models is widely adopted, enabling high-performance models for tasks such as thermal runaway warning with reduced data and computational requirements [157,158]. The trained models are then distilled into lightweight versions through quantization and pruning, making them suitable for real-time inference on vehicle-grade edge devices [159].

The system's dynamic intelligence relies on efficient cloud-edge collaboration. At the edge, lightweight onboard models perform millisecond-level inference to provide instant detection and rapid response to thermal runaway risks. Edge devices conduct localized processing of signals including current, voltage, temperature, ambient conditions, road status, humidity, and geographic information [160]. In the cloud, continuously uploaded key summaries support large-scale model retraining and validation, enabling ongoing model evolution. Updated model capabilities are deployed across the fleet via OTA, forming a closed-loop in which data-driven model improvement enhances edge performance. This collaborative architecture integrates operational data with simulation results to enable comprehensive battery state assessment and safety early warning through big data analytics. The entire process based on the Large Model-Driven early warning system is shown in Figure 6.

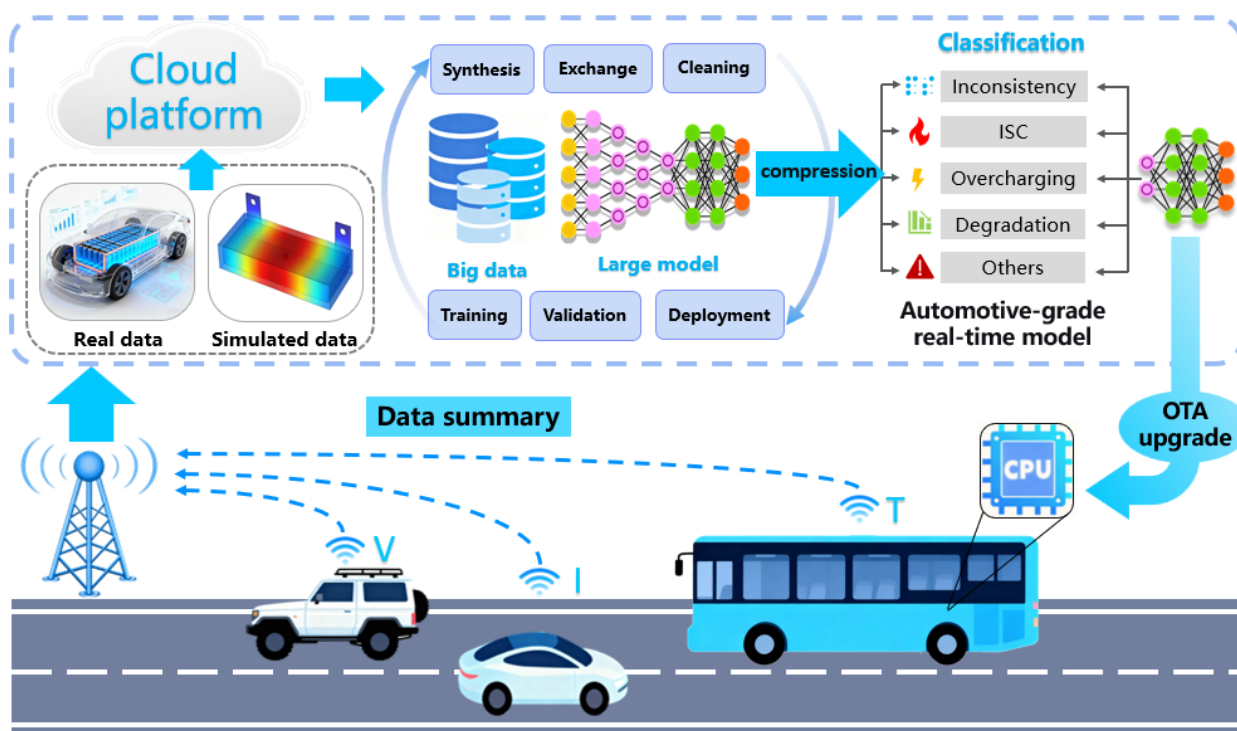


Figure 6. Next-generation intelligent early warning systems empowered by large models.

Zhao et al. [161] systematically reviewed deep learning methods for lithium-ion battery health prediction under multi-physical and multi-scale conditions, organized publicly available battery datasets, and discussed key techniques including Transformer-based architectures and transfer learning. Their reported results indicate that large-scale pre-trained models can achieve approximately 5% to 15% improvements in F1 scores in cross-domain transfer tasks compared with conventional machine learning baselines. This quantitative improvement suggests enhanced robustness against variations caused by battery aging and operating condition shifts, supporting the practical advantages of large model-driven approaches in complex application scenarios.

Despite the transformative potential of large model-driven methods, their industrial deployment faces three critical hurdles that require further benchmarking. First, data scarcity remains a primary bottleneck; high-quality, labeled thermal runaway datasets are rare due to the destructive and costly nature of abuse testing. While strategies like transfer learning and few-shot learning [162,163] show promise, model robustness under rare failure modes remains to be rigorously validated. Second, computational constraints on edge BMS chips often conflict with the heavy inference requirements of large models. Although model quantization and pruning [164] can reduce complexity, deploying high-parameter models without compromising the millisecond-level response time required for safety-critical protection remains a significant challenge [165]. Third, data privacy and security concerns arise when transmitting high-fidelity battery data to cloud platforms. To address this, recent studies have proposed federated learning frameworks that allow collaborative model training without exposing raw sensitive user data [166]. Consequently, future research must establish standardized benchmarks that evaluate not only prediction accuracy but also inference latency, data privacy compliance, and robustness under data-sparse conditions.

3.4. Summary of Prediction Strategies

Sections 3.1–3.3 have extensively detailed the operational principles of individual detection methodologies. However, the practical selection of an early-warning strategy for battery management systems necessitates a rigorous trade-off analysis between performance metrics and implementation constraints. Signal-based methods, while currently dominating industrial applications due to their high maturity and cost-effectiveness, are often limited by the physical latency of signal propagation such as heat conduction time or sensor susceptibility to environmental noise. Model-based approaches offer a theoretical advantage by estimating internal states that are not directly measurable, yet their real-time deployment is frequently hindered by the heavy computational burden and the difficulty of accurate parameter identification over the battery lifecycle. In contrast, emerging data-driven techniques demonstrate superior capabilities in capturing complex and non-linear fault patterns while extending warning lead times significantly, although they currently grapple with challenges related to data scarcity and model interpretability.

To provide a holistic decision-making framework, Table 4 systematically synthesizes and compares these three primary prediction categories. This quantitative summary highlights the distinct differences in critical indicators including lead time, detection accuracy, industrial maturity, and deployment costs. The aim is to guide researchers and engineers in selecting the most appropriate strategies tailored to specific application scenarios ranging from cost-sensitive electric vehicles to safety-critical stationary energy storage systems.

Table 4. Performance assessment and trade-off analysis of mainstream early-warning strategies.

Feature/Metric	Signal-Based (Voltage, Temp, Gas)	Model-Based (ECM, Electrochemical)	Data-Driven (ML, Big Data)
Typical Lead Time	Varies	Real-time Estimation	Predictive
Detection Accuracy	Moderate to High	High	Very High
Industrial Maturity	High	Medium	Low
Deployment Cost	Low to High	Low	High
Primary Advantages	Direct measurement; High reliability; Simple implementation	Internal state visibility; Mechanism-based	Handles complex non-linearities; Proactive life-cycle warning
Key Limitations	Single-source blind spots; Thermal lag (Temp); Sensor stability issues (Gas)	High computational load; Parameter identification difficulty	Heavy dependence on high-quality labeled data; Poor interpretability

4. Thermal Runaway Suppression Method

4.1. Thermal Management for Suppressing Early Heat Accumulation

The lithium-ion battery thermal management system serves as an indispensable safeguard for ensuring battery safety, capable of effective intervention against early-stage heat generation. A variety of thermal management strategies have been developed, which can be broadly categorized into active cooling, passive cooling, and hybrid cooling, as illustrated in Figure 7.

(1) Air cooling system

Air convection and nitrogen suppression are the main forms of air-cooling BTMS [167]. As shown in Figure 7a, Airflow can moderate pack temperature and lower gas concentration and explosion risk during thermal runaway [168], while pack arrangement and airflow design strongly influence cooling performance [169–172]. Bidirectional-flow structures further reduce peak temperature from 42.3 °C to 33.1 °C [76]. Nitrogen cooling can dilute oxygen and delay combustion [173]. Air cooling is simple and inexpensive but limited by low thermal conductivity and weak universality.

(2) Liquid cooling system

Liquid cooling provides higher heat transfer capability via direct or indirect contact, offering better temperature uniformity [174]. As depicted in Figure 7b, Silicone oil immersion achieves both rapid cooling and oxygen isolation with pack temperature controlled within 17.5–32.8 °C [175]. Water and nanofluid coolants show modest improvements [176], while optimized flow channels and microchannel plates significantly lower maximum temperature and temperature gradients (<2–5 °C) [177–180]. Liquid cooling offers excellent uniformity but increases system weight and power consumption, reducing usable SOC [181].

(3) Phase change material

Phase change material (PCM) can absorb large amounts of heat and offer low cost and high stability [182]. As shown in Figure 7c, Coupling PCMs with thermoelectric devices can maintain $\Delta T < 5$ K and long cooling duration [183]. PCM coatings improve heat transfer [184], and numerical studies confirm reduced peak temperatures and gradients [185]. Composite PCMs using graphene, copper foam, or expanded graphite greatly enhance thermal conductivity and reduce battery temperature rise by 5–20 °C [186–190]. PCMs have strong lightweight potential and are often combined with air or liquid cooling for improved performance [191].

(4) Heat pipe system

Heat pipes enable passive, high-conductivity heat transfer using phase-change cycling [192]. As illustrated in Figure 7d, They offer low maintenance and high efficiency in EV BTMS [193]. Experiments show heat pipes maintain pack temperatures below 50 °C and temperature differences under 5 °C at moderate heat loads [194]. CFD-validated designs reveal that structural parameters, especially conductive-element height, strongly affect thermal distribution, achieving 27.6 °C maximum and 1.1 °C gradient [195]. Heat pipes also support cooling and rapid heating (−20 °C to 0 °C in ~500 s) under extreme conditions [196]. Integration complexity and cost remain challenges.

(5) Thermoelectric cooler system

Thermoelectric coolers (TECs) enable precise, zonal temperature control using the Peltier effect [197]. As shown in Figure 7e, Advanced control strategies such as spectral modeling and NMPC improve temperature regulation accuracy [198]. Cold plate temperature significantly affects uniformity [199], while TEC-integrated copper-plate systems outperform forced air cooling and reduce cell temperature by 4–5 °C with gradients below 3 °C [175]. However, TEC efficiency is low and energy consumption is high, so TECs are generally auxiliary components [200].

(6) Hybrid cooling system

Hybrid systems combine air, liquid, PCM, and heat pipes to exploit complementary advantages. As summarized in Figure 7f, Liquid–air systems can maintain <33.6 °C and <5 °C gradients under 4C discharge [201]. PCM–liquid systems limit peak temperature to 39.4 °C and delay thermal-propagation during runaway [190]. Heat-pipe–liquid systems keep pack temperatures <40 °C with improved uniformity [202]. Heat-pipe–PCM split systems further lower peak temperatures by 7.6 °C compared with conventional PCM designs [203]. Hybrid cooling thus provides robust thermal control under complex duty cycles.

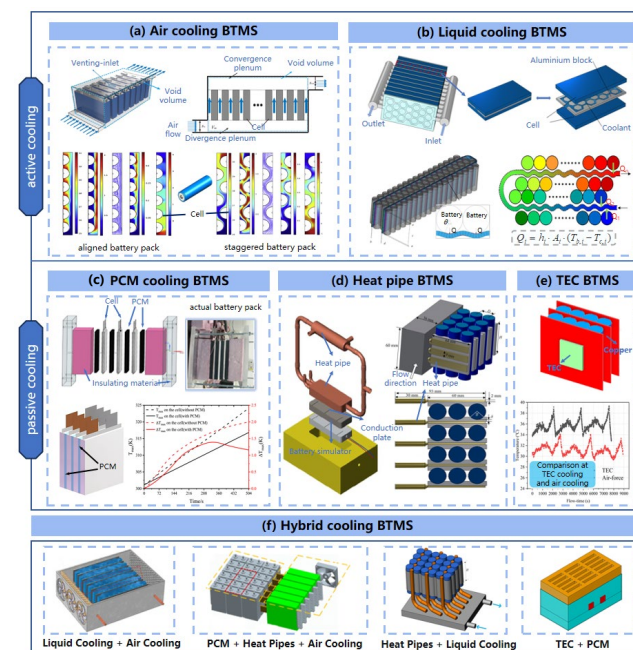


Figure 7. Based on thermal management system to suppress early heat accumulation. (a) Air cooling BTMS [166,167]; (b) Liquid cooling BTMS [177,186]; (c) PCM cooling BTMS [178,185]; (d) Heat pipe BTMS [192,195]; (e) TEC BTMS [197]; (f) Hybrid cooling BTMS, including liquid cooling + air cooling, PCM + heat pipes + air cooling, heat pipes + liquid cooling, and TEC + PCM [183,201–203].

4.2. Material-Based Suppression of Side Reactions

Material-level optimization can effectively suppress mid-term parasitic reactions, enhance cycling life, and improve long-term stability of lithium-ion batteries.

4.2.1. Positive Electrode Improvements

Surface coatings such as phosphates, fluorides and oxides enhance cathode thermal stability by isolating active materials from electrolytes and reducing heat generation from interfacial side reactions [204–206]. Synergistic bulk surface strategies, such as boron doping combined with $\text{Li}_2\text{B}_4\text{O}_7$ coating for NCM811, effectively suppress oxygen release and interfacial decomposition while enhancing lithium-ion transport [207]. FePO_4 coatings greatly improve overcharge resistance [206], whereas AlF_3 coatings increase the thermal runaway onset temperature by nearly 20 °C. Element substitution or doping (Ni, Cu, etc.) improves lattice stability, diffusion kinetics and thermal behavior of layered and spinel cathodes [208–210].

4.2.2. Negative Electrode Improvements

Negative electrodes produce substantial heat once the SEI decomposes at elevated temperatures. Stabilizing the SEI through polymer or inorganic coatings helps suppress dendrite growth, improve cycling efficiency and reduce charge transfer resistance [211–216]. Mild oxidation can form dense oxide layers that enhance thermal stability and suppress exothermic reactions. Artificial SEI films based on inorganic oxides or polymers improve rate capability, high-temperature endurance and can even act as thermal shut-down barriers during overheating.

4.2.3. Electrolyte Improvements

Electrolyte flammability and high reactivity accelerate heat and gas generation during abuse. Safety enhancements include flame retardants, overcharge additives, thermally stable lithium salts, polymer or inorganic solid electrolytes, and ionic liquid or thermosensitive formulations [217,218]. Alternative salts such as lithium difluoro phosphate and imide-based lithium salts offer significantly improved thermal stability and resistance to decomposition compared with conventional LiPF_6 [219,220]. Additives like DMAc improve high-temperature cycling and stabilize SEI compositions on both anode and cathode surfaces [221]. Chelated borate salts exhibit high decomposition thresholds (approximately 293 °C) [222]. Additives including EC, FEC, phosphates and fluorinated carbonates suppress SEI breakdown, reduce oxygen release and lower overall heat generation [223–226].

4.2.4. Separator Improvements

High-temperature-resistant separators reduce shrinkage and delay internal short-circuit formation. Polyimide aerogel membranes provide excellent ionic conductivity and structural stability at temperatures up to 120 °C [227]. Multilayer PP/PE/PP or PET membranes offer added thermal robustness [228,229], while ceramic and silica coatings further enhance heat resistance [230,231]. Thermoplastic polyurethane separators can react with PF_5 -derived species to form insulating layers under overheating, effectively isolating oxygen and improving flame retardancy [232].

4.2.5. Synergistic Effects and Full-Cell Performance

While single-component modifications are effective, the systemic safety of full cells is ultimately dictated by the intrinsic boundaries of the least stable constituent [233]. Wu et al. [234] found that the combined use of a thermally stable cathode and a functionalized electrolyte yields a safety margin superior to linear superposition. Similarly, Zhu et al. [235] leveraged the superior interfacial compatibility between composite sep-

arators and high-flash-point electrolytes, with both strategies significantly elevating the thermal runaway triggering threshold and suppressing peak heat release rates. Furthermore, advanced structural designs, such as the microencapsulation of flame retardants within separators [236], effectively resolve the trade-off between flame retardancy efficiency and kinetics. In summary, shifting from single-point breakthroughs to system-level synergy is crucial for reconciling the safety–performance trade-off in high-energy-density batteries.

4.3. Structural Strategies for Suppressing Late-Stage Deterioration

Structural protection focuses on responding to intrinsic signals such as pressure rise and resistance changes, enabling timely intervention to prevent propagation and shift battery safety from passive protection toward proactive early warning. This approach relies on advanced material design and component engineering to inhibit mid-term reaction rates and enhance intrinsic stability, as illustrated in Figure 8.

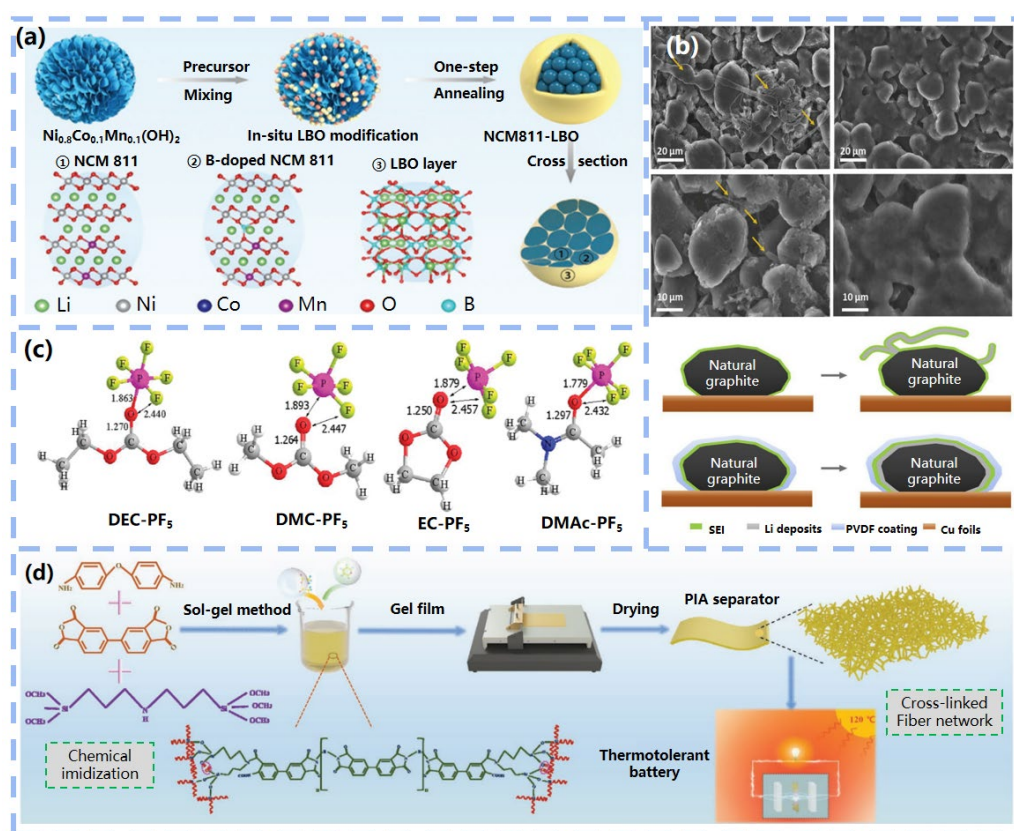


Figure 8. Inhibition of mid-term reaction rate based on battery materials. (a) Schematic illustration for the preparation of NCM811-LBO [204]. (b) Under over-lithiation conditions, the PVDF coating effectively suppresses lithium dendrite growth and promotes uniform lithium deposition [211]. (c) PF₅-solvent complex electrolyte geometry [221]. (d) Schematic illustration of the PIA separator [227].

4.3.1. Positive Temperature Coefficient (PTC) Thermistor

PTC conductive polymers provide current-limiting protection through sharp resistance increases near 100 °C [237–239]. Heating causes polymer expansion and separation of conductive particles, switching the device to a high-resistance state that reduces current load. Lyu et al. [240] demonstrated that TPU/PE/carbon composites produce a pronounced PTC jump near 120 °C, improving LiFePO₄ battery safety and maintaining cycling stability under thermal stress, as shown in Figure 9a.

4.3.2. Safety Vent

During thermal runaway, rapid heat and gas generation sharply increase internal pressure; safety vents act as designed weak points to release pressure and delay catastrophic rupture [241]. Vent-driven airflow can even power self-triggered warning devices without external energy [242]. Vent-opening pressure (P_{sv}) strongly affects thermal runaway severity: optimized P_{sv} (~ 0.35 MPa) lowers peak temperature and reduces hazard levels, whereas overly high P_{sv} risks internal overpressure [243], as shown in Figure 9b. In cylindrical cells, safety vents typically actuate above 100°C and at pressures higher than CID thresholds [244], while pouch cells rely on sealant fracture strength for controlled venting [245].

4.3.3. Current Interruption Device (CID)

CID disconnects the battery circuit when internal pressure or temperature reaches critical levels, preventing further reaction escalation [246,247]. Mechanical deformation such as alloy creeps may trigger premature CID activation and reduce available safety margins [248]. High-pressure CID activation in series-connected cylindrical cells can also cause voltage imbalances and arcing risks [249]. In Figure 9c,d, we have presented the different types of safety valves for batteries. Pressure-responsive CIDs rely on metal-foil rupture, whereas temperature-responsive CIDs use low-melting-point thermal fuses operating around $85\text{--}120^\circ\text{C}$. Since CID activation is irreversible, shape-memory-alloy-based reversible designs can mitigate permanent battery damage [250].

In Figure 9e, we present the arrangement patterns of various structures within the 18650 cylindrical battery, as well as the gas guidance mechanism when abnormal gas production occurs in the battery.

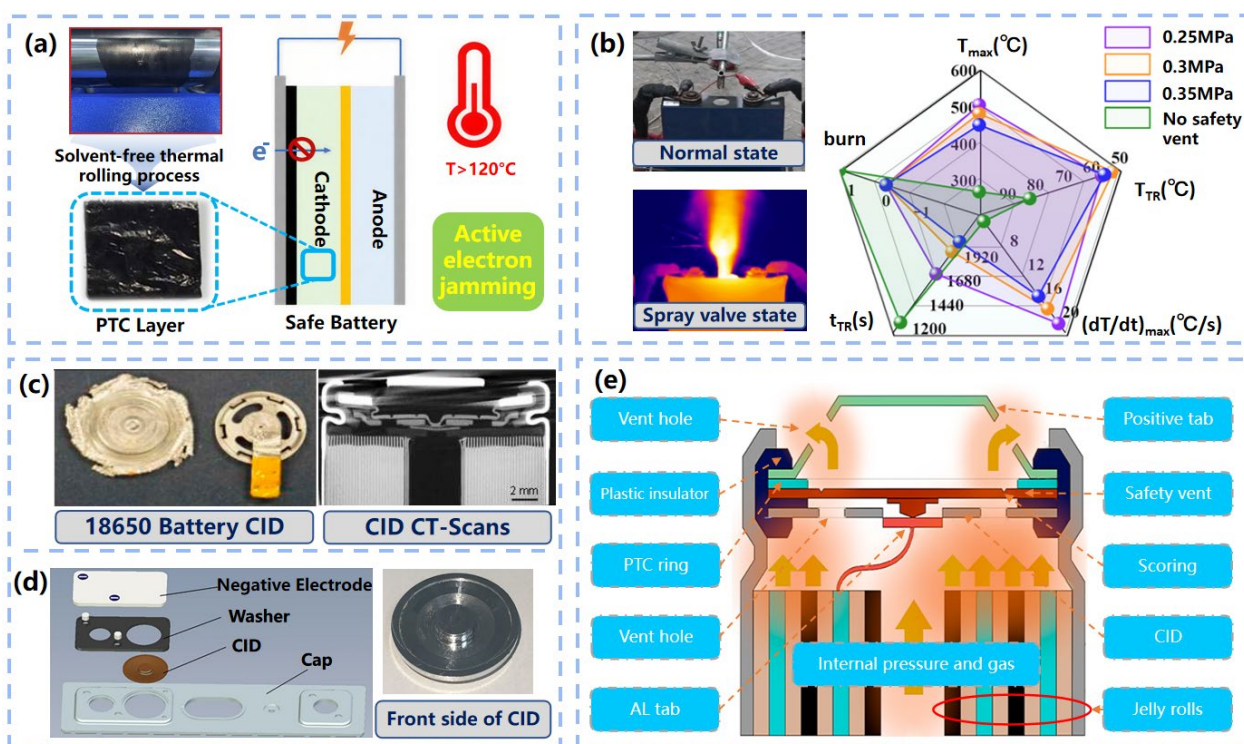


Figure 9. Thermal runaway suppression via battery design. (a) PTC Principle [240]; (b) Safety Vent: With vs. Without [242,243]; (c) CID in Prismatic Cells [246]; (d) CID in Cylindrical Cells [248]; (e) 18650 Battery Cross-section.

5. Future Perspectives and Outlook

Despite significant progress in understanding thermal runaway mechanisms and developing mitigation strategies, a critical gap remains between laboratory research and industrial deployment. Future research should focus on three prioritized phases to realize intrinsically safe energy storage systems.

The immediate priority is the standardization of abuse testing and evaluation protocols. Currently, the lack of unified testing standards constitutes a major barrier, resulting in poor comparability across studies. Future protocols must establish standardized methodologies and datasets encompassing realistic failure triggers rather than extreme, idealized conditions, ensuring high inter-laboratory consistency in recording critical safety parameters.

Algorithmically, research should emphasize integrating physics-informed intelligence with cloud-edge synergy to address the generalization and interpretability challenges of data-driven methods. Electrochemical principles must be embedded into neural network loss functions to ensure the physical plausibility of predictions even under data-sparse conditions. Efforts should center on “cloud-training, edge-inference” architectures to achieve minimal false alarm rates while maintaining high recall rates for thermal runaway events. Additionally, overcoming data privacy barriers via federated learning to enable collaborative model training without exposing raw data is a key research focus.

From the battery perspective, the ultimate goal is to eliminate thermal runaway through material innovation and intrinsic safety design. Research must address interfacial impedance and dendrite growth in solid-state electrolytes to develop components that remain non-flammable even at elevated temperatures. Simultaneously, developing “self-immunizing” materials is a promising direction, such as separators that terminate ion transport prior to thermal runaway onset or current collectors that drastically increase resistance upon short-circuiting. The vision is to achieve cell-level non-propagation, ensuring that a single-cell failure never escalates into a pack-level fire.

6. Conclusions

With the rapid expansion of lithium-ion battery applications, ensuring their safety has become an urgent scientific and engineering challenge. Thermal runaway remains the most critical failure mode, and understanding its evolution—initiated when heat generation persistently exceeds heat dissipation—provides the foundation for prediction and suppression strategies. Once the internal temperature rises, coupled electrochemical, thermal, and mechanical reactions accelerate, forming a positive feedback loop that eventually triggers catastrophic failure.

This review summarizes the current progress in the mechanism, prediction and early warning, and suppression strategies for thermal runaway. Existing early-warning approaches primarily rely on external sensing or internal state estimation. Single-signal monitoring offers high specificity and fast response, yet is inherently vulnerable to noise, sensor degradation, and multi-physics coupling effects. Consequently, multi-dimensional perception and fusion-based frameworks have emerged, enabling cross-verified risk assessment and substantially improving early-warning accuracy and lead time through collaborative analysis of heterogeneous signals. State-estimation-based methods further strengthen predictive capability by inferring internal reaction states beyond what external signals alone can capture.

Further improving battery safety requires a deeper, more systematic understanding of thermal runaway evolution under diverse fault and operating conditions. Key priorities include revealing the universal and fault-specific characteristics of thermal runaway, establishing quantitative correlations between multi-physics parameters and runaway risks, and developing predictive models with higher universality, robustness, and interpretability. On

the mitigation side, advances in high-efficiency thermal management, intrinsically stable materials, and intelligent structural protection will remain essential for balancing safety with high energy density and long-term durability. These research directions will collectively support the development of next-generation lithium-ion batteries with substantially improved intrinsic safety.

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